REVA: Region-Evidence Arbitration with Vocabulary-Adaptive Background Synthesis for Plain-Vocabulary Robustness in Training-free Open-Vocabulary Segmentation

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Abstract

Training-free open-vocabulary semantic segmentation (OVSS) methods lose 12–21 mIoU when the carefully engineered benchmark vocabulary is replaced by the plain class names a real user would type. We present REVA, a training-free, plug-in inference procedure that recovers this gap for *arbitrary* plain vocabularies. We scope the robustness claim to the plain-naming axis; the audit's other axes (synonyms, adversarial naming) are measured rather than claimed. REVA combines two components: (i) *vocabulary-adaptive background synthesis* (VABS), which selects a covering set of scene/stuff negative queries for the user's vocabulary via facility-location optimization with a similarity safety filter, and (ii) *region-evidence arbitration*, which averages dense per-pixel class probabilities inside SAM automasks and re-assigns each region by pooled evidence, with pixel-level fallback. On PASCAL VOC-21 (validation split excluding our development images), REVA lifts four training-free methods (MaskCLIP, SCLIP, ClearCLIP, NACLIP) from plain-vocabulary baselines by +19.9 to +23.1 mIoU, closing 86–96% of the gap to a matched-compute engineered-vocabulary upper bound (official vocabulary with the same region arbitration), and adds +4 to +9 mIoU on top of the official vocabulary itself. Every design decision was pre-registered with kill criteria and matched controls: the selection advantage over matched random negatives is positive in all 6 seeded cells on the full evaluation split (3 seeds, mean +3.1/+3.6 under region arbitration on SCLIP/NACLIP, range +1.2 to +5.1), and region arbitration passes the pre-registered no-added-harm criterion over pixel-level VABS (no class drops more than 3 IoU on two or more methods; worst single-method regression −3.2). External validity is anchored on unmodified official releases: VABS transfers to four official codebases (ProxyCLIP, SC-CLIP, Trident, CorrCLIP), and on SC-CLIP the full pipeline recovers 93% of the author-code naming-engineering gap. We report the boundaries equally: a pre-registered self-attack shows a tuned scalar background boost reproduces 73–85% of VABS's text-side gain (VABS's residual, untuned advantage is +2.8 to +4.4); the engineered vocabulary retains a 0.9–3.6 mIoU advantage under matched compute; SAM dominates runtime ($\sim$12× the CLIP pass); on Context-60 the negative-selection advantage vanishes; and a hand-written VOC background expansion matches VABS on VOC while losing to it off-VOC. The contribution is thus automation and vocabulary robustness, not a claim of a superior mechanism: REVA approaches hand-engineered-vocabulary quality with zero manual vocabulary work.

1 Introduction

Training-free OVSS methods [1–4] adapt CLIP [7] to dense prediction and are evaluated with fixed benchmark vocabularies. A controlled audit [9] showed that these official vocabularies embed undocumented engineering—background query expansion and positive sub-class expansion—worth

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11.8–20.6 mIoU on VOC-21, and that no pre-registered *text-only* mechanism could recover the engineered value safely: synthesised background negatives recovered the macro gap but systematically harmed classes such as *person* and *tvmonitor*, because harmful negatives are textually distant yet visually close, which no text-similarity filter can detect.

This paper provides the constructive answer that the audit predicted would require *independent visual evidence*. We arbitrate the competition between target classes and synthesised negatives at the level of visual regions rather than pixels: class-probability evidence is averaged inside SAM [8] automasks, and each region is assigned by pooled evidence. An oracle diagnostic motivates the design: pooling over ground-truth regions separates previously harmed target regions from background regions with AUC 0.92–0.97 under the same vocabulary in which pixel-level evidence failed, and the failure of a superpixel variant (SLIC) shows that region *quality*, not pooling itself, is the operative ingredient.

Our contributions:

1. **REVA**, a training-free plug-in that makes arbitrary plain vocabularies usable: vocabulary-adaptive background synthesis (facility-location selection over a scene/stuff lexicon with a similarity safety filter) plus SAM-region probability pooling with pixel fallback. No fine-tuning, no SAM decoder re-prompting, no mask-discovery heuristics.

2. **A pre-registered evidence chain** with matched controls: every stage was frozen before test runs, with kill criteria; two of our own stages (SLIC pooling and a DINOv2 visual veto) were killed by their criteria and are reported as negative results.

3. **Honest boundary mapping**: matched random negatives isolate the selection advantage; under matched compute the engineered vocabulary keeps a 0.9–3.6 mIoU lead (REVA closes 86–96% of the gap, it does not eliminate it); a hand-written-background control shows REVA *matches* manual engineering on VOC-21 rather than exceeding it (while beating the same list transplanted off-VOC); a LaVG-style feature-pooling variant performs on par with probability pooling; against the plain baseline, person and tvmonitor remain harmed (bounded, disclosed per class); and on Context-60 the selection advantage disappears.

2 Related Work

Training-free OVSS. Trained open-vocabulary segmenters align masks to language with segmentation or image-text supervision [39–43]; the training-free line instead re-reads a frozen CLIP [7]. MaskCLIP [1] re-projects CLIP's value features; SCLIP [2] and ClearCLIP [3] modify final-block attention; NACLIP [4] adds neighbourhood priors. REVA is a plug-in on top of any of them.

SAM/visual evidence in OVSS. Prior work couples SAM with CLIP either by training fusion modules (SAM-CLIP [15], OVSAM [16], ESC-Net [17]), by prompt-based re-decoding of coarse predictions (Trident [13]), or by ensembling automasks with CRF outputs (CaR [10]); LaVG [12] grounds text on unsupervised Ncut masks via feature-prototype similarity, and ProxyCLIP [5] injects SAM/DINO features as attention proxies. REVA instead performs output-level arbitration by averaging dense per-pixel class *probabilities* inside SAM automasks with pixel-level fallback, leaving encoders and the base method untouched. We do not claim novelty for region pooling itself—our ablation shows LaVG-style feature pooling performs comparably on SAM masks—but for the combination with vocabulary-adaptive negatives and its effect on vocabulary robustness.

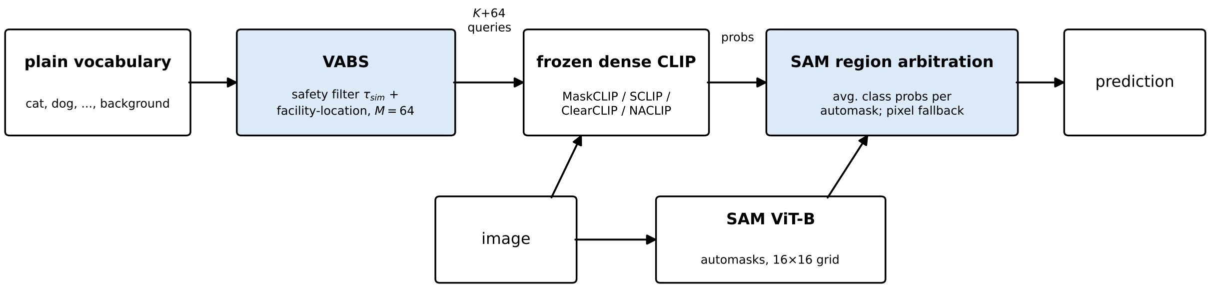

Figure 1: REVA overview. Both components are training-free and leave the base method untouched: VABS synthesises vocabulary-conditioned background negatives (blue); SAM automasks arbitrate target/negative competition by pooled probability evidence (blue). Pixels covered by no mask keep their pixel-level prediction.

Background modelling. Hand-written background query lists are folklore (SCLIP, CaR, LaVG); OVDiff [11] synthesises visual negative prototypes with a diffusion model; TCC [18] derives test-time contrastive concepts for single-concept segmentation. VABS synthesises a negative set for arbitrary *multi-class* plain vocabularies by facility-location coverage optimization over a fixed scene/stuff lexicon with a similarity safety filter. A same-protocol anchor against Trident's SAM-refinement stage and PAMR is reported in Table 3; a TCC comparison remains future work (§5).

Vocabulary robustness. On the classification side, prompt and name phrasing are long known to shift zero-shot accuracy [7, 36–38]. For segmentation, the audit [9] established the protocol and the plain-vs-official gap that REVA closes. Image-conditional vocabulary *pruning* (FreeCP [20]; concurrently ActiveSAM [22] on the SAM 3 concept-prompt stack) subtracts classes to reduce false-positive competition, whereas VABS *adds* negatives to repair background under-modelling — orthogonal interventions on the inference vocabulary (our own detector-guided pruning candidate, §4.6, is of the former kind). Concurrent SynCLIP [25] obtains synonym robustness by synonym-coherent re-pretraining of the encoders; REVA is a pure inference-time plug-in and targets a gap (plain-vs-engineered background modelling) that re-pretraining does not address. Concurrent PTC [24] calibrates the *text* embeddings of the given classes with visual prototypes, modulating calibration strength per image and class by evidence amount; it introduces no background negatives and no label-free applicability signal (its dataset-level dial is tuned with labels, nearly closing on COCO-Object with μ=0.02 vs 0.15–0.35 elsewhere), and it reports no switch-off rule for the harmful regime it observes there — independently corroborating the low-headroom boundary we characterise, while intervening on an orthogonal quantity.

3 Method

Setting. A frozen training-free OVSS method produces dense patch embeddings aligned to CLIP text space. The user supplies K plain class names (including a bare "background"). Text queries use the standard 80-template mean embedding [7]; multi-word entries joined by commas act as sub-queries of one class with max-pooled probabilities (SCLIP convention).

VABS: vocabulary-adaptive background synthesis. Given target embeddings $T \in \mathbb{R}^{K\times d}$ and a scene/stuff lexicon L (COCO-Stuff [30] ∪ ADE-150 [31] class names, 168 entries after splitting and de-duplication), VABS (i) removes candidates with $\max_k \cos(t_c, T_k) \geq \tau_{\text{sim}} = 0.90$ (safety filter), then (ii) greedily selects $M = 64$ negatives by facility-location coverage [35] of the remaining candidate space with a redundancy penalty. Selected negatives become sub-queries of the background class: any pixel they win is background. Hyper-parameters (M, τ_{sim}) were fixed on a 100-image development split (disjoint from all test images) in earlier pre-registered work and never re-tuned.

Why discrete negatives and per-patch max. Two pre-registered probes fix the aggregation choice mechanistically. Replacing the discrete negative set by a continuous background *subspace* (projection-norm energy of the patch feature onto the top-32 singular directions of the lexicon embeddings, scale-anchored per image) collapses to the plain baseline (34.5/34.4 vs 53.6/52.7 for the discrete set on SCLIP/NACLIP, VOC test-300): a global energy scalar is nearly uniform across patches and carries no per-patch discrimination. A follow-up sweep interpolating between the two — background score = mean of the top-k negative probabilities, $k \in \{1, 2, 4, 8, 16, 64\}$ — degrades *monotonically* with k on all three hosts and both datasets (e.g. SCLIP VOC 53.6 $\rightarrow$ 52.7 $\rightarrow$ 50.7 $\rightarrow$ 47.0 $\rightarrow$ 40.5 $\rightarrow$ 27.2; k=64, the discrete analogue of the energy score, falls below plain), and the winning negative is patch-dependent beyond a handful of words: on 50 VOC images per host, the eight globally most frequent winners cover only 58–64% of background-won patches; and a follow-up pre-registered pruning test shows the coverage is not the whole story — keeping only the global top-24 winners (which cover $\sim$92% of wins on the VOC diagnostic images) still loses against the full set by 0.58–1.12 mIoU in five of six host–dataset cells (the sixth loses 0.46, just inside the frozen 0.5 bar), so lower-ranked negatives contribute through runner-up competition beyond their win counts. The background channel therefore needs per-patch discrete competition against the full diverse set; every averaging- or pruning-based alternative we froze in advance performs worse (single seed, shared CLIP backbone across hosts; run records in the artifact).

Region-evidence arbitration. SAM (ViT-B, automatic mask generation, 16×16 point grid) proposes class-agnostic masks on the resized image. Masks are rasterised largest-first so smaller masks overwrite larger ones, preserving detail; pixels covered by no mask keep their pixel-level prediction. For each region r the class-probability map (after softmax with logit scale 40 and per-class sub-query max pooling) is averaged: $p_r = \frac{1}{|r|} \sum_{x \in r} p(x)$, and the region is assigned $\arg\max_c p_{r,c}$.

Why regions, not confidence. Our earlier text-only attempt failed because dense CLIP confidence is diffuse: at threshold 0.5, zero high-confidence *person* patches exist, so no pixel-level filter can protect harmed classes. The oracle diagnostic (§4.1) shows evidence pooled over true regions is separable, i.e. information the pixel view cannot expose.

4 Pre-registered evidence chain

All experiments run on OpenCLIP [32] ViT-B/16 (OpenAI weights), short side 336, window 224, stride 112, logit scale 40, no post-processing (no PAMR [6], no multi-scale/flip). Benchmarks are PASCAL VOC-21 [27], PASCAL-Context-60 [28], COCO-Object [29], and ADE-150 [31]. This minimal unified protocol yields lower absolute numbers than published method-specific pipelines (e.g. our NACLIP official-vocabulary baseline is 56.9 versus the published 64.1, which uses PAMR refinement — resolution settings match ours); all claims in this paper are *within-protocol deltas* under identical settings across arms, not state-of-the-art absolute comparisons. Development used 100 VOC images (validation images 300–399); all VOC test numbers exclude these 100 images

Table 1: VOC-21 validation split excluding the 100 development images (1349 images; standard all-class mIoU from a globally aggregated confusion matrix). "pix" = pixel-level, "+SAM" = region-evidence arbitration. rand. 64 = matched random negatives (single seed here; a 3-seed full-split replication gives a seeded selection advantage of +3.1/+3.6 mean, positive in all cells; finding (ii) below). REVA = plain + VABS + SAM. Official numbers use the engineered benchmark vocabulary; official+SAM is the matched-compute upper bound. The audit's image-resampling noise floor on 300-image subsets is 3.1 mIoU; this full-split table is less noisy, but sub-mIoU differences are not interpreted.

	plain vocabulary					official vocabulary	
Method	pix	pix+SAM	pix+VABS	rand.64+SAM	**REVA**	pix	+SAM
SCLIP	35.6	36.6	55.5	53.6	**58.7**	57.2	61.2
ClearCLIP	34.8	35.2	52.9	50.2	**56.1**	55.7	59.7
MaskCLIP	32.8	39.0	45.1	47.5	**52.7**	44.6	53.6
NACLIP	36.9	39.6	54.1	54.5	**59.2**	56.9	62.6

(1349 images). Because every class occurs in the ground truth of the test images, the reported mIoU is the standard all-class dataset-level mIoU computed from a globally aggregated confusion matrix—not a per-image GT-present average, avoiding the metric artifact documented in the audit [9]. Criteria were frozen before each test run; we report the chain including its kills. Throughout the paper, gains and deltas quoted in the text are computed from unrounded run records; they may differ from differences of the displayed values (rounded to 0.1) by up to 0.1.

4.1 Stage 0: oracle diagnostic and the SLIC kill

Pooling raw similarities over ground-truth connected components separates previously-harmed target regions (person, tvmonitor) from background regions with AUC 0.92/0.97 under the VABS vocabulary—the same vocabulary under which pixel-level evidence harmed those classes. But the same pooling over SLIC superpixels *decreased* dev mIoU by 3.0, killing the superpixel variant by its pre-registered criterion and isolating region quality as the bottleneck.

4.2 Stage 1: SAM regions (main result)

Replacing SLIC with SAM automasks passed all frozen criteria on dev and was then validated on the full split (Table 1).

Findings (dev-excluded split): (i) SAM arbitration alone barely helps the plain vocabulary (+0.4 to +6.2; column pix+SAM)—VABS and arbitration are only strong *together*: region arbitration adds +3.2 to +7.6 over pixel-level VABS, for a total plain→REVA gain of +19.9 to +23.1; (ii) the selection advantage over matched random negatives is +4.7 to +5.9 under region arbitration (Table 1, REVA vs. rand.64+SAM; single seed here); a pre-registered 3-seed replication of the random-negative set on the *full* dev-excluded split (1349 images, region arbitration) gives seeded advantages of +5.1/+1.2/+2.9 (SCLIP, mean +3.1) and +4.7/+2.1/+4.0 (NACLIP, mean +3.6): positive in all 6 cells. An earlier test-300 replication of the same seeds was noisier (−0.1 to +3.8, spread comparable to the 3.1 subset noise floor), so subset-level estimates of this advantage should carry that caveat; (iii) REVA on plain vocabularies exceeds the official vocabulary evaluated *at pixel level* for all four methods—but this is an unequal-compute comparison: under matched compute (official + SAM) the engineered vocabulary retains a 0.9–3.6 mIoU lead, i.e. REVA closes 86–96% of the engineered-vocabulary gap rather than eliminating it (per method:

SCLIP 90%, ClearCLIP 86%, MaskCLIP 96%, NACLIP 87%); (iv) the plug-in is orthogonal: applied on top of the official vocabulary it still adds +4.0 to +9.0; (iv-b) a pre-registered self-attack control (the companion audit's one-parameter test: any absorption claim must beat a direct background calibration): even when the competitor is handed a labelled advantage — a *single scalar background-logit boost* tuned on a 100-image labelled dev split, where VABS gets no per-vocabulary tuning — VABS still wins by 2.8–4.4 under the frozen criterion, and the boost adds only +0.2 to +1.1 on top of VABS. The tuned boost does reproduce 73–85% of the text-side VABS gain on VOC (SCLIP 34.7 → 50.7 vs VABS 53.5; NACLIP 36.6 → 48.4 vs 52.8), and it is brittle (the best boost differs per model and +0.08 collapses SCLIP to 33). A pre-registered transfer check settles whether the boost is a portable competitor: the VOC-tuned scalar *harms* on every transfer cell (Context-60 −0.3 to −0.9 on plain, −5.8 to −9.6 on top of VABS; ADE-150 −1.3 to −1.4, where the plain protocol has no background class to boost at all) — the boost is a dataset-specific calibration constant requiring labelled re-tuning per benchmark, whereas untuned VABS gains on Context-60 and only reproduces its already-disclosed ADE null. We read VABS as automatic, vocabulary-conditioned background re-calibration plus a genuine selection component, and report the boost as a baseline any negative-vocabulary method should be compared against; (v) per-class safety under the pre-registered criterion (REVA vs. pixel-level VABS, i.e. does arbitration *add* harm): no class drops more than 3 IoU on two or more methods (single worst case: person −3.2 on NACLIP). Against the *plain* baseline, however, the harm inherited from text-only negative synthesis is reduced but not eliminated: person remains −6.0/−6.0/+8.6/−8.9 and tvmonitor −2.9/−11.2/−6.3/−11.8 (SCLIP/ClearCLIP/MaskCLIP/NACLIP; Table 4). The same signature extends to — and is somewhat larger on — the three 2024–25 style-reimplementation bases (test-300): person −7.7/−15.4/−13.0 and tvmonitor −13.9/−12.0/−13.0 on ProxyCLIP- [5]/LPOSS-[19]/SC-CLIP-style [26] respectively, with LPOSS tvmonitor dropping to near zero (2.1), while pottedplant improves by +13 to +18 on all three. REVA trades bounded losses on 2 of 21 classes for a +20-level macro gain; users for whom person or tvmonitor is critical should not adopt the plug-in blindly. A pre-registered attempt to *predict* the at-risk classes from text alone fails: the max text cosine of a class to the 64 selected negatives does not rank per-class deltas (Spearman +0.37/+0.39, wrong sign vs. the frozen hypothesis, $p>0.09$; full split, 20 classes) — so the per-class safety table, not a text-side screen, remains the deployment check.

4.3 Stage 2: visual veto (killed)

A pre-registered second stage re-assigned background-won regions whose DINOv2 [34] embedding matched a high-confidence same-image target anchor. Best dev gain over stage 1 was −0.08 mIoU (criterion: $\geq +1$); the stage is dead. Region pooling already captures the recoverable signal; remaining background regions similar to target anchors are predominantly true background.

4.4 Transfer and boundary conditions

On a COCO-Object protocol (80 things + background, full val2017), the full pipeline lifts the plain-vocabulary baseline from 22.4/22.4 to 34.4/36.3 (SCLIP/NACLIP); region arbitration contributes +2.5/+3.9 over pixel VABS with a reduced selection advantage (+1.1/+0.8). On PASCAL-Context-60 (full split), pooling still adds +2.0/+3.1 over pixel VABS (31.3/31.9 → 33.3/35.0; plain 31.2/31.8) but VABS is indistinguishable from random negatives—consistent with the audit's finding that low-background-room regimes leave no space for negative selection. Across benchmarks the effect tracks the size of the background-modelling room the audit identified, as its boundary-condition model predicts: VOC-21 (26-name engineered background expansion) +19.9 to +23.1;

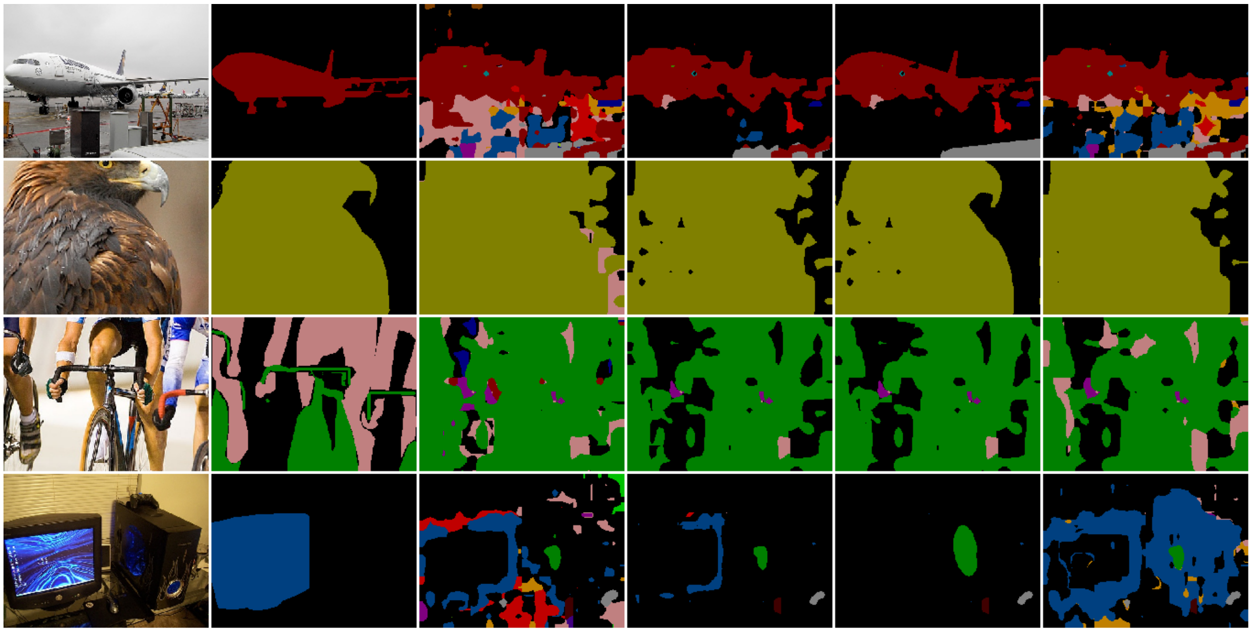

Figure 2: Qualitative results (NACLIP). Columns: image, GT, plain vocabulary (pixel), plain+VABS (pixel), REVA (VABS+SAM), official vocabulary (pixel). Rows 1–2: VABS suppresses plain-vocabulary false positives and SAM arbitration cleans region boundaries. Rows 3–4 are honest failure cases: the person cyclist is partly absorbed by negatives/bicycle, and the tvmonitor is lost by REVA while the official vocabulary's engineered queries recover it—the residual per-class harm quantified in Table 4.

COCO-Object (all stuff remapped to background) +12.0/+13.9; Context-60 (59 classes already cover the scene) +2.0/+3.1; ADE-150 (no background channel at all) +1.5/+1.9. REVA is not a one-benchmark phenomenon — it is a repair whose available gain is set by how much background under-modelling the target protocol has. We retain the earlier disclosure that the scene lexicon shares names with COCO-Stuff, which the COCO-Object protocol remaps to background: part of the negative set coincides by construction with ground-truth background semantics on this benchmark, so the COCO-Object transfer overstates what a fully leak-free lexicon would achieve. Two further pre-registered transfers sharpen the boundary. On ADE-150 (150 classes, no background class; full val, 2000 images), region arbitration still helps but shrinks to +1.5/+1.9 over pixel VABS (SCLIP 16.31→17.83, NACLIP 17.26→19.14), and VABS itself is null: no gain over plain names (−0.35/−0.35) and indistinguishable from random negatives (−0.10/+0.02)—without a background class to absorb into, vocabulary-adaptive negative selection has no room, the same mechanism as Context-60. On an OpenCLIP ViT-L/14 backbone (VOC-21 dev-excluded full split, 1349 images), the full six-cell grid on SCLIP and NACLIP shows the pipeline transfers: SCLIP plain 37.7 → REVA 45.9 and NACLIP plain 35.7 → REVA 53.1, in both cases *above* the pixel-level official reference (42.0 / 51.6; no matched official+SAM bound was run at ViT-L, and ViT-L dense baselines are globally weaker than ViT-B). The VABS-vs-random selection advantage under arbitration is +4.0 on NACLIP and +2.5 on SCLIP at full split (an earlier test-300 cell had SCLIP at +1.8, below our +2 replication bar; single seed); VABS negatives were selected with ViT-B text embeddings and reused unchanged. We therefore scope the selection-advantage claim to background-rich vocabularies, note that it is method-dependent at ViT-L scale, and observe that region arbitration transfers in every regime tested.

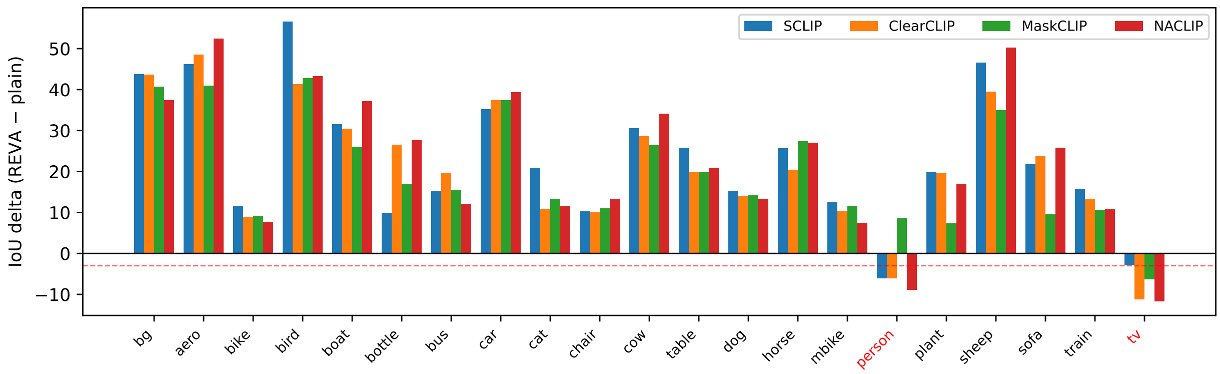

Figure 3: Per-class IoU change of REVA relative to the plain-vocabulary pixel baseline (dev-excluded VOC). Every class improves on every method except person (down on three of four methods; MaskCLIP +8.6) and tvmonitor (down on all four) — the two classes the audit identified as victims of text-only negative synthesis (red labels); the dashed line marks −3 IoU.

Finally, REVA transfers beyond the attention-surgery family. Applied to a ProxyCLIP-style base (DINO proxy attention over CLIP values, our style-reimplementation of a 2024 visually-guided method; VOC-21 test-300), plain vocabulary rises from 37.8 to 58.5 — closing 99% of the gap to the pixel-level engineered-vocabulary reference (58.8), or 85% of the gap to the matched official+SAM bound (62.1), on a fifth base method whose dense mechanism differs qualitatively from the other four — with a +4.2 advantage over matched random negatives. An external anchor on the *unmodified* official ProxyCLIP release (authors' protocol, only the class-name file swapped) confirms the text-side component outside our stack: adding the VABS-64 negatives to the plain name file lifts author-code mIoU from 47.3 to 56.2, recovering 64% of the author-code naming-engineering gap (official name file: 61.2); the SAM arbitration component cannot be attached without modifying their code and is therefore not anchored. A second author-code anchor with a pre-registered random control, the unmodified official SC-CLIP release (full VOC val): VABS-64 lifts plain from 43.0 to 59.6 (+16.7, recovering 77% of its author-code naming-engineering gap to 64.6), while the same budget of matched random negatives reaches 56.5 — a +3.1 selection advantage on author code (single run, single negative seed), consistent with the +3.1/+3.6 full-split means measured in our own stack (Table 1). On this release the SAM arbitration component is anchored as well, applied as a pre-registered *additive* layer on the official probability output (the official forward is untouched; SAM ViT-B, our frozen config; the repo's own background threshold re-applied identically): full REVA (VABS+SAM) reaches 63.2 from plain 43.0, recovering 93% of the author-code naming-engineering gap to the *pixel-level* official reference (64.6; a matched official+SAM bound was not run on official code, so this percentage is not compute-matched, unlike the 85–96% in-stack closures), with a +3.5 arbitration gain (59.62→63.16) over pixel-level VABS and a +4.2 advantage over SAM-arbitrated matched random negatives (58.96); plain+SAM alone gains only +1.1 (44.1), so the arbitration gain is contingent on the recalibrated background, exactly as in our stack. For ProxyCLIP, only the VABS component remains anchored. A third author-code anchor tests VABS *inside* a SAM-coupled pipeline: on the unmodified official Trident release (CLIP+DINO splicing with SAM-encoder aggregation and SAM refinement, full VOC val, only the class-name file changed), VABS-64 lifts plain from 47.4 to 63.5 (+16.0, recovering 82% of its author-code naming-engineering gap to 67.1, which reproduces the published value), with a +2.8 advantage over matched random negatives (60.7) — so the selection advantage survives

Table 2: Author-code dose curve (unmodified official Trident and CorrCLIP, full val, only the class-name file changed): the naming-engineering headroom (NEG) determines what negative expansion can do. All values are mIoU; selection is VABS − matched random; parenthesised selection values are differences between two harmful arms and are reported descriptively only, with no benefit claim.

Host / dataset	NEG	VABS − plain	selection	regime
Trident / VOC-21	+19.7	+16.0	+2.8	recovery + selection
CorrCLIP / VOC-21	+20.5	+15.2	+3.9	recovery + selection
Trident / COCO-Object	+2.0	+1.3	−0.1	parity, no selection
CorrCLIP / COCO-Object	+1.4	−4.1	(−0.8)	harmful
Trident / Context-60	+0.2	−3.4	(+0.7)	harmful

even where the host method already exploits SAM internally. This covers Trident, previously absent from our comparisons, as a host *for the VABS component only* (pixel-level name-file folding; Trident's own SAM refinement retained as shipped; our region arbitration was not applied on this host). Extending the same anchor to COCO-Object (author-code naming-engineering gap only +3.3): *negative expansion* of either kind reaches official parity there — VABS 37.9 and matched random negatives 38.0 from plain 34.4 — so the pre-registered selection-advantage criterion *fails* (−0.0) and the recovery cannot be credited to vocabulary-conditioned selection. Rather than a bare failure, this is a third independent confirmation of the gap-vs-recovery boundary (after the in-stack dataset gradient and the below-bar LPOSS cell): where the background headroom is small, any negative expansion suffices, and the selection advantage is a large-headroom (VOC-like) phenomenon on author code as in our stack. A fourth, pre-registered confirmation comes from official Trident on COCO-Object (its author-code naming gap is only +2.0): there even the VABS *gain* criterion fails (+1.3, below our +3 bar) and VABS is indistinguishable from matched random negatives (40.36 vs 40.43 from plain 39.1; selection −0.1) — the boundary prediction was stated in the preregistration before the run and held. Where the audit finds no background headroom, REVA has nothing to recover; users should consult the audit's per-dataset NEG before deploying it. The zero-headroom end is confirmed on official Trident Context-60 (author-code NEG only +0.2): there negative expansion actively *harms* — VABS −3.4 and matched random −4.1 against plain — reproducing on author code the null-to-harmful ADE boundary we disclose in-stack; we make no benefit claim from the +0.7 VABS-vs-random difference between two harmful arms. A fourth official codebase, CorrCLIP [14] (ICCV 2025 oral; SAM2 region masks + MetaCLIP + DINO, the current training-free performance ceiling), reproduces the recovery end of the curve and *extends* its harmful end, under the same pre-registered thresholds (single run, single negative seed, as for the other author-code anchors; the repository's region masks are pre-generated and identical across arms by construction): on VOC (author-code NEG +20.5) VABS-64 lifts plain from 54.3 to 69.5 (+15.2, recovering 74% of the gap) with a +3.9 advantage over matched random negatives (65.6); on COCO-Object (NEG only +1.4) both expansions actively harm (VABS −4.1, random −3.3 from plain 42.3). The frozen prediction (VABS − plain below the +3 bar at near-zero headroom) held, and this extends the author-code harmful regime up to a NEG of +1.4 — the largest headroom at which harm has been observed. Note the cross-host ordering tension this exposes: CorrCLIP harms at NEG +1.4 while Trident merely ties at +2.0, so NEG does not monotonically order regimes *across* hosts — consistent with our treatment of the composition heuristic as an observation, not a validated predictor (§5). As on Trident, this covers CorrCLIP for the VABS component only. Table 2 summarises the author-code dose curve.

The companion audit's generational extension shows this base is exactly as naming-sensitive as the surgery family (NEG +22.1 on the full dev-excluded split), so the repair is as relevant for the newest generation as for the old. The same protocol on two further 2024–25 style-reimplementations brings the total to seven base methods: on SC-CLIP-style anomaly restoration, REVA lifts plain from 38.1 to 58.5 (above its engineered-vocabulary pixel reference of 56.7, VABS-vs-random +3.1); on LPOSS-style label propagation it lifts plain from 41.2 to 57.4 (94% gap closure), but the VABS-vs-random advantage there is only +1.3, below our +2 replication bar — label propagation partially absorbs the distinction between negative vocabularies, so on that family we claim gap closure from region arbitration but not a negative-selection advantage. Against the *matched* official+SAM upper bounds, which we have now run for all three bases (ProxyCLIP 62.1, LPOSS 60.3, SC-CLIP 60.9 on the same test-300 protocol), the gap closures are 85.4%, 85.2% and 89.4% respectively; SC-CLIP REVA (58.5) exceeds its pixel-level official reference (56.7) but remains 2.4 below the matched bound. The SC-CLIP figure of +3.1 is the VABS-vs-random selection advantage (58.5 vs 55.4), not the margin over its official reference.

4.5 Controls mandated by novelty checking

Two same-protocol controls on the dev-excluded split (1349 images): (i) *LaVG-style feature pooling* (average dense features in the region, then cosine argmax) scores 59.2/57.3/52.7/59.3 (SCLIP, ClearCLIP, MaskCLIP, NACLIP) versus our probability pooling 58.7/56.1/52.7/59.2: feature pooling is equal or slightly better (up to +1.2 on ClearCLIP), so we claim no superiority for probability pooling; its role here is only that it operates on calibrated class probabilities, which the VABS negatives require. (ii) *Hand-written background control* (SCLIP's official 26-entry background expansion on the plain vocabulary, same SAM pooling) scores 59.0/58.4/53.0/61.0: manual engineering is equal or slightly *better* (up to +2.3 on ClearCLIP)—VABS's value is automation for vocabularies where no hand-engineered expansion exists, not superior quality. A pre-registered off-VOC check tests whether the adaptivity ever pays: transplanting the same 26-entry list verbatim to vocabularies it was not written for, VABS beats it on COCO-Object by +1.3/+2.2 (30.48/30.65 vs 31.81/32.81 GT-present, SCLIP/NACLIP, pixel arms, test-300), while on ADE-150 both are null-to-slightly harmful (the disclosed ADE boundary extends to the hand list). The static list is thus a VOC-shaped artifact; where its assumptions break, the vocabulary-conditioned selection is what remains (single seed; both arms share partial stuff→background lexicon overlap on COCO-Object).

What the selection advantage is, mechanically. A pre-registered accounting on test-300 (region arbitration, matched random negatives) locates the advantage: VABS raises GT-background *recall* by +5.0/+4.7 points (SCLIP 89.2 vs 84.2; NACLIP 91.2 vs 86.5) at no precision cost (+1.3/+0.8). At pixel level, a VABS negative wins 33–36% of pixels versus 17–19% for random negatives: vocabulary-conditioned negatives are competitive against foreground classes where random words are not, so background stops leaking into objects. The advantage is background recall, not foreground discrimination — consistent with the boost self-attack's finding that most (not all) of the text-side gain is background re-calibration.

Perturbed-vocabulary axes. A pre-registered check runs REVA on the audit's other two axes (VOC-21 test-300, VABS negatives regenerated by the frozen recipe conditioned on each perturbed vocabulary; baseline arm = SAM-region pooling on the bare perturbed vocabulary). Under 100% synonym substitution REVA lifts SCLIP 32.1 → 49.9 (+17.8) and NACLIP 31.6 → 46.8 (+15.2), retaining 85–88% of its plain-regime gains (+20.4/+17.9): the mechanism survives synonym perturbation, but the residual synonym cost (≈7 mIoU below the plain-vocabulary REVA level) is not repaired — REVA restores background modelling, not naming quality. On an adversarially

Arbitration	Vocabulary	SCLIP	NACLIP	s/img
none (pixel)	plain+VABS	55.5	54.1	–
PAMR	plain+VABS	–	58.5	0.09
Trident SAM-refinement	plain+VABS	58.4	**59.8**	0.13
REVA (SAM automask pooling)	plain+VABS	**58.7**	59.2	0.42
none (pixel)	official	57.2	56.9	–
PAMR	official	–	62.6	0.08
Trident SAM-refinement	official	61.0	**63.0**	0.13
SAM automask pooling	official	**61.2**	62.6	0.42

Table 3: External anchor mechanisms on the dev-excluded VOC-21 split (1349 images), all on the same dense probabilities and (for SAM rows) the same SAM ViT-B checkpoint. Timing is the refinement cost on top of the CLIP pass (0.035 s). PAMR was run on NACLIP only.

searched (ANS) vocabulary REVA lifts SCLIP 15.3 → 25.1 and NACLIP 14.7 → 23.9: a real but partial recovery, reported as a stress bound — REVA is not a defense against worst-case naming. Single seed, single substitution seed (s0). The ANS vocabulary was searched (by the audit) on the first 100 images of the split, which are contained in the 300-image evaluation subset used here, so the ANS cell is an in-search-domain stress bound rather than a held-out attack.

External anchors. Two reviewer-requested anchor rows on the same split: (i) *PAMR* [6], the standard affinity-based refinement used by published pipelines, applied to our NACLIP pixel probabilities: official vocabulary 56.9 → 62.6 and VABS vocabulary 54.1 → 58.5, at 0.08 s/image. On the official vocabulary PAMR *matches* our SAM arbitration (62.6) at a fraction of the cost; with VABS negatives SAM arbitration retains a +0.7 edge (59.2 vs 58.5). We report this honestly: for engineered vocabularies cheap affinity refinement suffices, and the specific value of region arbitration is concentrated in the synthesized-negative regime it was designed for. It also explains most of the reproduction gap to published NACLIP (62.6 vs 64.1; the remainder is attributable to residual pipeline differences of the official repository — its resolution settings match ours, so the gap is not a resolution artifact). (ii) *Trident-style SAM refinement* [13]: their released prompt-based refinement stage (connected components of the coarse prediction prompt the SAM decoder with points, boxes and mask priors) ported onto our dense probabilities with the same SAM ViT-B checkpoint; results in Table 3. The two SAM-based mechanisms are close on both vocabularies (within 0.4 mIoU on SCLIP in our favour, within 0.7 against us on NACLIP), and the prompt-based decoder pass is ∼3× cheaper than automask generation. We therefore do *not* claim our probability-pooling arbitration is a superior SAM-usage mechanism; the vocabulary contribution (VABS) is orthogonal to and composable with either. Full Trident additionally splices CLIP/DINO features and uses SAM ViT-H at higher resolution; our row reproduces only its refinement stage under our protocol and is an anchor, not a reproduction of their published numbers.

4.6 A third component candidate: detector-guided vocabulary pruning

This section reports a *candidate* extension under evaluation; it is not part of REVA proper, and no headline claim of this paper (abstract, contributions, main tables) includes it. A pre-registered follow-up probe tests whether an external open-vocabulary detector's *box support* can prune the dense argmax: a vocabulary entry is kept only if OWLv2 (base-ensemble, untuned threshold 0.2) fires at least one box for it in the image; background is always kept. Across nine cells ({SCLIP, ClearCLIP, NACLIP} × {VOC-21, Context-60, COCO-Object}, 300-image subsets) pruning improves the dense model by +1.4 to +9.2 mIoU (mean +4.4), harms no cell, and does not amplify naming fragility (the synonym arm also gains; the synonym drop is unchanged, 4.8 vs 4.2). It composes additively with REVA on all three methods tested (VOC): SCLIP 57.0 → 59.4, ClearCLIP 54.5 → 57.0, NACLIP 57.5 → 61.1 — on this 300-image subset the SCLIP cell reaches 59.4 against an engineered-vocabulary pixel reference of 58.8 (not a main-table claim; the main tables use the full dev-excluded split). A pixel-flow accounting suggests why the synonym arm also gains: 55% of the foreground pixels that synonym renaming leaks away go to classes *absent* from the image (only 3% to present classes), which is exactly the mass pruning removes; we report this descriptively, as an aggregate accounting with no class-level claims — a class-level graded correlation between absent-leak fraction and pruning gain did not survive pre-registered testing (Spearman −0.20). Two pre-registered boundary probes: on ADE-150 the gain persists but *shrinks* (+2.5 to +2.7 for SCLIP/NACLIP vs +6 to +9 on VOC) — it does not grow with vocabulary size as an absent-mass account would predict — and a pre-registered granularity probe rules out the obvious explanation: OWLv2's per-class presence recall does *not* degrade with WordNet [33] label depth (Spearman +0.17, wrong sign) and does not predict per-class pruning gain (+0.05); mean presence recall on ADE is 0.67, i.e. a third of present classes are wrongly pruned yet the net gain survives. A bounded pixel ledger locates the shrinkage on the loss side: the correction mass (dense-wrong pixels rescued by pruning) is nearly constant across VOC and ADE, but the destruction mass (dense-correct pixels whose GT class was pruned) grows 2.6–3.5× as the mispruned present-class share rises from 14% to 34% — and the mispruning is fault-tolerant because the detector tends to miss exactly the classes dense CLIP already segments poorly (dense IoU 10–30 for mispruned vs 28–50 for kept classes), so deletions cost less than their count suggests; *why* detector misses concentrate there remains open (a pre-registered soft variant that down-weights unsupported classes by λ=0.3 instead of deleting them recovers +0.6 of the ADE loss but costs up to 1.7 on VOC — a mild, not free, mitigation); a matched-budget trivial baseline — CLIP image-level top-k filtering with k set per image to the number of classes the detector keeps — reaches ∼80% of the gain on VOC but is *net negative* on ADE-150 (−3.9 vs +2.5 to +2.7 for detector pruning), so the detector evidence, not generic class filtering, is what keeps pruning safe as the vocabulary grows; and under the searched worst-case vocabulary of the companion audit, pruning lifts the attacked arm by ∼7 mIoU but recovers only ∼35% of the damage — it is not a defense against attacks that degrade the scoring of *present* classes. Honest framing: this adds a second model at inference (an OWLv2 pass per image, a cost of the same order as the SAM stage that already dominates our runtime), it re-imports the detector's own vocabulary sensitivity into the pipeline (the companion audit measures a synonym drop of 7.1 for OWLv2 — mild, though not a family property: Grounding DINO drops 37.6 under the same suite, so the detector choice matters — and REVA with pruning is no longer a pure CLIP+SAM plugin), and that model is *grounding-supervised* with a pretraining distribution that covers the benchmark classes — pruning imports external supervision rather than extending the training-free recipe, consistent with the companion audit's finding that per-class protection tracks supervision; the standalone detector pipeline remains stronger on detection-friendly classes, so the claim is strictly dense-vs-dense-pruned; and this is a preliminary single-seed result on 300-image subsets — we report it as a third-component *candidate*, not an established component. Notably, the same signal fails completely as a distractor-axis shield (see the companion audit): the detector also fires on near-distractor names; its usefulness here is pruning absent classes from clean vocabularies, not detecting vocabulary pollution.

4.7 A fourth component candidate: corpus-statistics negative selection with a derived budget

Like §4.6, this reports a *candidate* under evaluation, outside REVA proper and outside all headline claims. VABS selects negatives by facility-location coverage of the *text* embedding space; the per-patch win-statistics diagnostic of §3 suggests an image-side alternative: run the host once over 50 *unlabelled* images from the target domain, count how often each candidate negative wins a background patch, and keep the most-used negatives. Both the winner list and the budget are derived from the statistics alone (closest prior art selects *templates* for positive classes from unlabelled-corpus entropy statistics, FLOSS [21], or prunes *foreground* vocabularies per image [20]; concurrent TPA [23] runs the host over a small unlabelled deployment pool and aggregates confident patches into *positive visual* prototypes — the same unlabelled-deployment-statistics skeleton, but on the positive/visual side, with no negative words and no derived budget; to our knowledge no prior work selects background *negatives* from unlabelled target-domain statistics) — M^* is the smallest budget covering 90% of background wins — replacing the free budget parameter by a single frozen coverage constant (90%, fixed before any run; a pre-registered sweep over 80–95% moves the result by at most 0.8 mIoU on VOC and never breaks the low-headroom no-harm bound on ADE-150). A pre-registered evaluation (three hosts $\times$ four datasets $\times$ three disjoint statistics windows, all disjoint from the evaluated images; matched facility-location controls at the same M^*) passed all frozen criteria: on VOC and COCO-Object the statistics-selected set beats the matched text-side control by +0.5 to +5.0 mIoU in all 18 cells; on the low-headroom Context-60/ADE-150 boundary it never harms beyond the frozen 0.3 tolerance (18/18 cells); the derived budget is stable across windows (within a factor 1.9) and adapts in the right direction, shrinking to 7–17 words on ADE-150 versus 19–34 elsewhere. The selected set uses roughly a third of the VABS-64 budget at a cost of 0–1.1 mIoU (on ADE-150 it beats the full set in 6/9 cells and is within 0.15 in the remaining three), and the selection transfers across stacks: plugged into the unmodified official ProxyCLIP release, statistics-selected negatives beat the matched text-side control at the same budget by +1.6 mIoU for the derived-budget selection (M^*=21, 59.2 vs 57.7) and +1.5 for the earlier fixed-24 registration, both pre-registered (run records in the artifact). Caveats: single evaluation seed, shared CLIP backbone across the three hosts, statistics windows drawn from the same benchmark image distribution as the evaluation split (disjoint images, same domain — the realistic deployment assumption; a pre-registered cross-domain probe, counting statistics on the *other* benchmark's images, still beats the matched control in all six cells by +1.8 to +4.1 mIoU, at 0–1.2 below the same-domain selection — both benchmarks are natural-image distributions, so only near-domain shift is covered and larger shifts remain untested); an earlier registration of this component failed one frozen gate by 0.02 mIoU and is reported in the artifact alongside this re-registration; the re-registration dropped that gate's plain-relative condition (a post-hoc verification finds it would have passed in all 36 cells here) and pre-committed a one-miss-per-dataset tolerance that the results did not need. A final pre-registered promotion gate (official ProxyCLIP, Context-60, plain vocabulary) *failed*: while the statistics selection again matches the text-side control (33.7 vs 34.0, within the 0.3 tolerance), *any* negative expansion harms this low-headroom cell relative to plain (36.5) by $\sim$2.8 mIoU — consistent with the applicability boundary of §4.4, and the reason this component remains a candidate outside REVA proper: it improves *which* negatives to use, not *whether* to use them.

5 Limitations

Same-protocol anchors now cover Trident's SAM-refinement stage and PAMR (Table 3); they show our arbitration is competitive but not uniformly superior, and a TCC comparison and full-Trident reproduction (CLIP/DINO splicing, SAM ViT-H, higher resolution) remain unrun, so our absolute numbers are still not comparable to published multi-scale pipelines (see §4). Single backbone (ViT-B/16); SAM adds an external visual model whose automatic mask generation dominates runtime: measured per-image cost is 0.42 s for SAM + pooling versus 0.035 s for the CLIP dense pass ($\sim 12\times$; RTX 3090, points-per-side 16). The matched random-negative control is now replicated with 3 seeds at full-split scale (all cells positive; Table 1); other controls remain single-seed as noted. The author-code anchors validate VABS on four official releases (ProxyCLIP, SC-CLIP, Trident, CorrCLIP) and full REVA (VABS+SAM, additive on the official probability output) on one (SC-CLIP, VOC); the arbitration layer has not been run as a *replacement* for any official repository's own postprocess, and the ProxyCLIP anchor covers VABS only. A pre-registered *additive* redundancy test on one further official release bounds where arbitration helps: applied on top of official CorrCLIP — whose shipped postprocess already mode-votes over its pre-generated SAM2 regions — our layer fails the pre-registered $\geq +1.0$ transfer criterion (VOC, single run: VABS arm 69.5 $\to$ 68.2, -1.3 vs the official postprocess; matched random arm 65.6 $\to$ 64.1, -1.6; both small negative deltas are reported descriptively, no harm claim), matching the pre-registered frozen prediction of failure/redundancy. This is consistent with arbitration adding value only where the host has not already aggregated region evidence — a single-dataset boundary observation on one host pair, not a validated host-level rule. OpenReview-stage collision checking was unavailable (provisional novelty). **Applicability warning**: the author-code dose curve (Trident VOC NEG +19.7 $\to$ VABS +16.0; Trident COCO-Object +2.0 $\to$ tie; CorrCLIP COCO-Object +1.4 $\to$ VABS *harms*, -4.1; Trident Context-60 +0.2 $\to$ VABS *harms*, -3.4) means REVA should not be deployed on low-headroom regimes — benchmarks whose vocabulary already densely covers scene content (stuff-inclusive label spaces where background is a minor residual class), as opposed to thing-centric vocabularies with a large residual background. This composition heuristic is consistent with every measured point but is a post-hoc observation, not a validated predictor; an automatic, deploy-time applicability test (the user cannot measure NEG without an engineered vocabulary) is an open problem. A pre-registered pilot on this problem found one label-free signal with the pre-declared sign: vocabulary *crowding* (mean max image–query cosine similarity on 50 unlabelled images) anti-correlates with the oracle expansion gain across 8 host–dataset configurations (Spearman -0.69 with primary-source gains; the two hosts share four datasets, so the effective sample is closer to 4 and the pass over the frozen 0.6 bar is thin); prediction entropy, despite correlating more strongly, has the *wrong* pre-declared sign (confounded by vocabulary size) and a frozen global-threshold switch failed its do-no-harm criterion (host-dependent signal scales), so we report a candidate signal, not a validated detector. A pre-registered ordinal out-of-sample check on the official codebases (read-only crowding margin from each repo's stored class map, 50 unlabelled images, single run) ordered 4 of 5 within-host dataset pairs correctly ($n = 5$ pairs), with the single miss coming from the repo with the deepest stored post-processing; by the frozen criterion this is marginal and carries no claim. A pre-registered expansion under a single matched extraction protocol (raw cosine, 18 points: three hosts $\times$ four datasets plus synonym-substituted vocabularies on two of them) passed both its pooled and per-host bars (pooled Spearman -0.72; -0.83 within every host — the host-scale confound that broke the global switch does not break the within-host ordering; the three hosts share the CLIP backbone and produce *identical* rank orderings, so this is one within-host test replicated three times, not three independent replications), but its vocabulary-axis test failed 0/6: within a fixed dataset, replacing class names by synonyms lowers both the crowding score and the expansion gain, the *opposite* of the frozen prediction. The signal is therefore a dataset-level headroom proxy that must be calibrated per host, not a vocabulary-level one; for synonym substitution it cannot see naming-induced headroom changes — a pre-registered robustness check over two additional random second-tier synonym draws replicated the inverted pattern in all 12 of 12 plain-vs-synonym pairs (three hosts $\times$ two datasets $\times$ two draws). A pre-registered ViT-L/14 replication did not clear its frozen bar (one host -0.89, the other -0.49 against a -0.5 bar, and the crowding ordering coincides with ViT-B/16's, so backbone independence remains unestablished); the signal stays a pilot. Development images come from VOC; cross-dataset hyper-parameter transfer beyond the reported protocols is untested. Finally, a pre-registered attempt to *amortise* REVA into a learned text-side adapter with universal background queries (trained on 8k unlabelled ADE images with REVA pseudo-labels and a naming-invariance loss) failed all its criteria: universal learned negatives cannot contain vocabulary-relative words (sky/grass are VOC negatives but training targets), and invariance training degraded held-out synonym vocabularies. The vocabulary-adaptivity of VABS at inference time appears essential, and REVA's inference cost cannot currently be trained away on the text side (records in the companion audit's artifact). A follow-up probe that moved the same naming-invariance objective to a visual-side adapter over frozen dense features also failed its pre-registered criteria (held-out synonym vocabularies degraded, the same failure signature), and a per-region configuration-routing oracle showed a +14.9 mIoU ceiling that no label-free gate we tested could access; both are reported in the companion audit as negative results; a follow-up router fit on 50 labelled images also failed to beat the best fixed configuration, so the routing ceiling remains inaccessible to every signal family we tested. ADE-150 and ViT-L/14 transfers (§4.4) bound the applicability of negative selection to background-rich vocabularies, with a method-dependent advantage at ViT-L scale.

6 Conclusion

REVA turns the audit's negative result into a largely constructive one: region evidence arbitration prevents the *additional* per-class harm that killed text-only negative synthesis, and recovers 86–96% of the engineered-vocabulary gap under matched compute, under pre-registered criteria and matched controls. What it does not do is also reported: the residual harm to person and tvmonitor relative to the plain baseline is reduced but not eliminated, the engineered vocabulary retains a 0.9–3.6 mIoU lead at matched compute, and SAM dominates inference cost. The per-class protective component of engineered vocabularies thus remains only partially automatable—a refined version of the audit's open problem.

A Per-class IoU on the dev-excluded VOC split

Table 4 reports the full per-class breakdown behind the safety criterion of §4.

B Run provenance

All numbers derive from JSON run records stored with the code (`runs/clean_*.json` for the dev-excluded VOC split, `runs/full_*.json` for the transfer protocols); every table cell is traceable to a specific record. Timing is recorded per run. The 100 development images (validation indices 300–399) are excluded from every VOC test number in this paper; earlier pre-registered dev decisions used only those images.

Table 4: Per-class IoU (%) on the dev-excluded VOC-21 split: plain-vocabulary pixel baseline (pl), pixel-level VABS (pix), and REVA (+SAM). The pre-registered safety criterion compares +SAM against pix (arbitration adds no harm; worst single-method regression person −3.2 on NACLIP). Against pl, the harm inherited from text-only negative synthesis persists for tvmonitor on all four methods and for person on three of four (MaskCLIP +8.6; bounded at −11.8).

	SCLIP			ClearCLIP			MaskCLIP			NACLIP		
Class	pl	pix	+SAM	pl	pix	+SAM	pl	pix	+SAM	pl	pix	+SAM
background	40.9	83.0	84.7	38.9	80.8	82.5	43.4	80.0	84.1	48.3	83.6	85.7
aeroplane	14.6	52.3	60.8	17.6	53.8	66.1	12.3	33.2	53.3	23.6	59.4	76.0
bicycle	17.5	28.9	29.0	21.5	30.5	30.4	20.8	28.3	30.0	26.1	33.5	33.7
bird	21.7	70.9	78.3	39.0	71.5	80.3	22.3	52.0	65.1	37.1	70.1	80.3
boat	7.9	34.6	39.5	8.9	34.5	39.3	6.2	21.4	32.3	9.4	39.0	46.5
bottle	55.2	57.0	65.1	42.3	59.0	68.9	33.7	37.4	50.6	38.9	55.9	66.6
bus	70.9	80.3	86.1	61.6	73.4	81.2	56.6	57.4	72.1	66.4	68.2	78.5
car	35.1	68.1	70.3	28.2	63.1	65.6	25.9	52.0	63.3	31.3	65.6	70.7
cat	64.5	82.2	85.4	62.3	74.7	73.1	68.7	74.1	81.9	73.3	81.2	84.8
chair	19.2	27.1	29.4	18.6	27.3	28.7	22.4	26.7	33.4	17.3	26.3	30.6
cow	46.6	73.7	77.2	41.9	67.1	70.6	46.8	61.8	73.3	42.7	69.2	76.8
diningtable	16.9	41.4	42.6	20.9	39.0	40.8	12.8	27.8	32.5	20.0	38.5	40.7
dog	67.7	79.4	82.9	62.5	74.5	76.4	60.5	67.0	74.8	67.7	76.1	80.9
horse	36.1	59.2	61.8	33.7	52.1	54.1	34.7	54.3	62.0	38.9	61.4	65.9
motorbike	51.8	61.9	64.3	55.6	61.7	65.9	42.3	47.2	53.9	56.3	58.7	63.7
person	44.2	39.0	38.2	52.3	48.2	46.3	24.7	34.2	33.3	48.2	42.5	39.3
pottedplant	17.6	35.8	37.4	9.9	28.2	29.6	21.4	28.1	28.7	9.5	25.4	26.4
sheep	23.5	66.9	70.1	16.8	51.8	56.2	38.5	63.6	73.5	18.2	61.3	68.4
sofa	25.8	43.3	47.5	23.9	44.4	47.6	34.8	37.2	44.4	27.2	47.5	53.0
train	39.8	54.6	55.6	40.8	52.3	53.9	37.0	44.1	47.7	46.2	54.6	57.0
tvmonitor	29.8	25.7	26.9	32.4	23.2	21.2	22.3	18.4	16.0	29.2	18.6	17.4

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